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Education

1992	B.S., Computer Science, University of Utah, Salt Lake City, UT
2000	Ph.D., Computer Science, University of Utah, Salt Lake City, UT
	Committee: Prof. Elaine Cohen (Chair), Prof. Chris Johnson , Prof. Richard Riesenfeld

Professional Positions

2020–present	Distinguished Research Scientist and Visualization Group Leader, Oak Ridge National Laboratory
2012–2020	Senior Research Scientist, Oak Ridge National Laboratory
2007–2012	Research Scientist, Oak Ridge National Laboratory

Journal Articles

- [1] Wang, Z., Moreland, K., Larsen, M., Kress, J., Childs, H., Pugmire, D.. “*Parallelize Over Data Particle Advection: Participation, Ping Pong Particles, and Overhead*,” *IEEE Transactions on Visualization and Computer Graphics* 2025 DOI: [10.1109/TVCG.2025.3557453](https://doi.org/10.1109/TVCG.2025.3557453).
- [2] Athawale, T., Wang, Z., Pugmire, D., Moreland, K., Gong, Q., Klasky, S., Johnson, C., Rosen, P.. “*Uncertainty Visualization of Critical Points of 2D Scalar Fields for Parametric and Non-parametric Probabilistic Models*,” *IEEE Transactions on Visualization and Computer Graphics* 2025 DOI: [10.1109/TVCG.2024.3456393](https://doi.org/10.1109/TVCG.2024.3456393).
- [3] Athawale, T., Triana, B., Kotha, T., Pugmire, D., Rosen, P.. “*A Comparative Study of the Perceptual Sensitivity of Topological Visualizations to Feature Variations*,” *IEEE Transactions on Visualization and Computer Graphics* 2024 DOI: [10.1109/TVCG.2023.3326592](https://doi.org/10.1109/TVCG.2023.3326592).
- [4] Athawale, T., Johnson, C., Sane, S., Pugmire, D.. “*Fiber Uncertainty Visualization for Bivariate Data With Parametric and Nonparametric Noise Models*,” *IEEE Transactions on Visualization and Computer Graphics* 2023 DOI: [10.1109/TVCG.2022.3209424](https://doi.org/10.1109/TVCG.2022.3209424).
- [5] Yenpure, A., Sane, S., Binyahib, R., Pugmire, D., Garth, C., Childs, H.. “*State-of-the-Art Report on Optimizing Particle Advection Performance*,” *Computer Graphics Forum* 2023 DOI: [10.1111/cgf.14858](https://doi.org/10.1111/cgf.14858).
- [6] Liang, X., Whitney, B., Chen, J., Wan, L., Liu, Q., Tao, D., Kress, J., Pugmire, D., Wolf, M., Podhorszki, N., et al.. “*MGARD+: Optimizing Multilevel Methods for Error-Bounded Scientific Data Reduction*,” *IEEE Transactions on Computers* 2022 DOI: [10.1109/TC.2021.3092201](https://doi.org/10.1109/TC.2021.3092201).

- [7] Gainaru, A., Wan, L., Wang, R., Suchyta, E., Chen, J., Podhorszki, N., Kress, J., Pugmire, D., Klasky, S.. “*Understanding the Impact of Data Staging for Coupled Scientific Workflows*,” *IEEE Transactions on Parallel and Distributed Systems* 2022 DOI: [10.1109/TPDS.2022.3179989](https://doi.org/10.1109/TPDS.2022.3179989).
- [8] Authors not available. “*Co-design center for exascale machine learning technologies (ExaLearn)*,” *The International Journal of High Performance Computing Applications* 2021
- [9] Authors not available. “*HyLiPoD: Parallel Particle Advection Via a Hybrid of Lifeline Scheduling and Parallelization-Over-Data*,” *Unknown Venue* 2021
- [10] Authors not available. “*Machine Learning-Based Autotuning for Parallel Particle Advection*,” *Unknown Venue* 2021
- [11] Authors not available. “*Minimizing development costs for efficient many-core visualization using MCD3*,” *Parallel Computing* 2021
- [12] Authors not available. “*Scalable Multigrid-based Hierarchical Scientific Data Refactoring on GPUs*,” *arXiv preprint arXiv:2105.12764* 2021
- [13] Authors not available. “*Spatial coupling of gyrokinetic simulations, a generalized scheme based on first-principles*,” *Physics of Plasmas* 2021
- [14] Authors not available. “*The Exascale Framework for High Fidelity coupled Simulations (EFFIS): Enabling whole device modeling in fusion science*,” *The International Journal of High Performance Computing Applications* 2021
- [15] Authors not available. “*A terminology for in situ visualization and analysis systems*,” *The International Journal of High Performance Computing Applications* 2020
- [16] Authors not available. “*Accelerating Multigrid-based Hierarchical Scientific Data Refactoring on GPUs*,” *arXiv preprint arXiv:2007.04457* 2020
- [17] Authors not available. “*ADIOS 2: The Adaptable Input Output System. A framework for high-performance data management*,” *SoftwareX* 2020
- [18] Authors not available. “*Estimating Lossy Compressibility of Scientific Data Using Deep Neural Networks*,” *IEEE Letters of the Computer Society* 2020
- [19] Authors not available. “*Extending the publish/subscribe abstraction for high-performance I/O and data management at extreme scale*,” *Bulletin of the IEEE Technical Committee on Data Engineering* 2020
- [20] Authors not available. “*Global adjoint tomography—model GLAD-M25*,” *Geophysical Journal International* 2020
- [21] Authors not available. “*MGARD+: Optimizing multilevel methods for error-bounded scientific data reduction*,” *arXiv preprint arXiv:2010.05872* 2020
- [22] Authors not available. “*Optimizing multi-grid based reduction for efficient scientific data management*,” *arXiv preprint arXiv:2010.05872* 2020
- [23] Authors not available. “*The Moving Target of Visualization Software for an Ever More Complex World*,” *Unknown Venue* 2020

- [24] Authors not available. “*The moving target of visualization software for an increasingly complex world,*” *Computers & graphics* 2020
- [25] Authors not available. “*Towards 1ULP evaluation of Daubechies Wavelets,*” *arXiv preprint arXiv:2005.05424* 2020
- [26] Authors not available. “*3.17 ECP ALPINE Algorithm Overview,*” *In Situ Visualization for Computational Science* 2019
- [27] Authors not available. “*4.3 Workflow Execution,*” *In Situ Visualization for Computational Science* 2019
- [28] Authors not available. “*4.5 Algorithmic Challenges,*” *In Situ Visualization for Computational Science* 2019
- [29] Authors not available. “*A New Generation of Earth Mantle Model from Global Adjoint Tomography,*” *Acta Geologica Sinica (Beijing)* 2019
- [30] Authors not available. “*XVis: Visualization for the Extreme-Scale Scientific Computation Ecosystem Final Report.,*” *Unknown Venue* 2019
- [31] Authors not available. “*Pillars of the Mantle,*” *Unknown Venue* 2017
- [32] Authors not available. “*Global adjoint tomography: first-generation model,*” *Geophysical Journal International* 2016
- [33] Authors not available. “*Preparing for in situ processing on upcoming leading-edge supercomputers,*” *Supercomputing frontiers and innovations* 2016
- [34] Authors not available. “*Vtk-m: Accelerating the visualization toolkit for massively threaded architectures,*” *IEEE computer graphics and applications* 2016
- [35] Authors not available. “*Visualizing the Radiation of the Kelvin-Helmholtz Instability,*” *IEEE Transactions on Plasma Science* 2014
- [36] Authors not available. “*A parallel EM algorithm for model-based clustering applied to the exploration of large spatio-temporal data,*” *Technometrics* 2013
- [37] Authors not available. “*Exploring Collaborative HPC Visualization Workflows using VisIt and Python,*” *Unknown Venue* 2013
- [38] Authors not available. “*GPU Acceleration of Particle Advection Workloads in a Parallel, Distributed Memory Setting,*” *Unknown Venue* 2013
- [39] Authors not available. “*The Ultra-scale Visualization Climate Data Analysis Tools (UV-CDAT): Data Analysis and Visualization for Geoscience Data,*” *Computer* 2013
- [40] Authors not available. “*Ultrascale visualization of climate data,*” *Computer* 2013
- [41] Authors not available. “*A distributed data-parallel framework for analysis and visualization algorithm development,*” *Unknown Venue* 2012
- [42] Authors not available. “*An examination of magnetized outflows from active galactic nuclei in galaxy clusters,*” *Monthly Notices of the Royal Astronomical Society* 2012

- [43] Authors not available. “*Applying Model-Based Clustering for Analysis of Community Atmospheric Model Output*,” *Unknown Venue* 2012
- [44] Authors not available. “*EAVL: the extreme-scale analysis and visualization library*,” *Unknown Venue* 2012
- [45] Authors not available. “*GPU-accelerated visualization*,” *High Performance Visualization* 2012
- [46] Authors not available. “*Oliver Rubel, Marc Durant, Jean M. Favre, and Paul Navratil. VisIt: An End-User Tool For Visualizing and Analyzing Very Large Data*,” *High Performance Visualization: Enabling Extreme-Scale Scientific Insight* 2012
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- [62] Authors not available. “4.8 Cost Models,” *In Situ Visualization for Computational Science* 1989
- [63] Authors not available. “3.11 EC’s Strategy towards Exascale,” *In Situ Visualization for Computational Science* Unknown
- [64] Authors not available. “Analysis of Recurrent Patterns in Toroidal Magnetic Fields,” *Unknown Venue* Unknown
- [65] Authors not available. “Compressible magma/mantle dynamics: 3-D, adaptive simulations in ASPECT,” *Unknown Venue* Unknown
- [66] Authors not available. “ECP VTK-m: Updating HPC Visualization Software for Exascale-Era Processors,” *Unknown Venue* Unknown
- [67] Authors not available. “Yearly Progress Report (May 2018): Plasma Surface Interactions: Predicting the Performance and Impact of Dynamic PFC Surfaces,” *Unknown Venue* Unknown

Conference Papers

- [68] Wang, X., Choi, J., Kurihaya, T., Lyngaas, I., Yoon, H., Xiao, X., Pugmire, D., Fan, M., Nafi, N., Tsaris, A., et al.. “ORBIT-2: Scaling Exascale Vision Foundation Models for Weather and Climate Downscaling,” *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis* 2025 DOI: [10.1145/3712285.3771989](https://doi.org/10.1145/3712285.3771989).
- [69] Senapati, P., Athawale, T., Pugmire, D., Guan, Q.. “Advancing Comprehension of Quantum Application Outputs: A Visualization Technique,” *Proceedings of the 2023 International Workshop on Quantum Classical Cooperative* 2023 DOI: [10.1145/3588983.3596689](https://doi.org/10.1145/3588983.3596689).
- [70] Bolstad, M., Moreland, K., Pugmire, D., Rogers, D., Lo, L., Geveci, B., Childs, H., Rizzi, S.. “VTK-m: Visualization for the Exascale Era and Beyond,” *ACM SIGGRAPH 2023 Talks* 2023 DOI: [10.1145/3587421.3595466](https://doi.org/10.1145/3587421.3595466).
- [71] Authors not available. “Error-controlled, progressive, and adaptable retrieval of scientific data with multilevel decomposition,” *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis* 2021
- [72] Authors not available. “Fides: A General Purpose Data Model Library for Streaming Data,” *International Conference on High Performance Computing* 2021
- [73] Authors not available. “Improving parallel particle advection performance with machine learning,” *Proc. Eurographics Symposium on Parallel Graphics and Visualization* 2021
- [74] Authors not available. “Neural data compression for physics plasma simulation,” *Neural Compression: From Information Theory to Applications–Workshop@ ICLR 2021* 2021
- [75] Authors not available. “Comparing Time-to-Solution for In Situ Visualization Paradigms at Scale,” *2020 IEEE 10th Symposium on Large Data Analysis and Visualization (LDAV)* 2020

- [76] Authors not available. “I/o performance characterization and prediction through machine learning on hpc systems,” *CUG2020 Proceedings* 2020
- [77] Authors not available. “Opportunities for cost savings with in-transit visualization,” *International Conference on High Performance Computing* 2020
- [78] Authors not available. “Parallel Particle Advection Bake-Off for Scientific Visualization Workloads,” *2020 IEEE International Conference on Cluster Computing (CLUSTER)* 2020
- [79] Authors not available. “Visualization as a Service for Scientific Data,” *Smoky Mountains Computational Sciences and Engineering Conference* 2020
- [80] Authors not available. “A lifeline-based approach for work requesting and parallel particle advection,” *2019 IEEE 9th Symposium on Large Data Analysis and Visualization (LDAV)* 2019
- [81] Authors not available. “Comparing the efficiency of in situ visualization paradigms at scale,” *International Conference on High Performance Computing* 2019
- [82] Authors not available. “In Situ Analysis and Visualization of Fusion Simulations: Lessons Learned,” *High Performance Computing: ISC High Performance 2018 International Workshops, Frankfurt/Main, Germany, June 28, 2018, Revised Selected Papers* 2019
- [83] Authors not available. “In situ particle advection via parallelizing over particles,” *Proceedings of the Workshop on In Situ Infrastructures for Enabling Extreme-Scale Analysis and Visualization* 2019
- [84] Authors not available. “PAVE: An In Situ Framework for Scientific Visualization and Machine Learning Coupling,” *2019 IEEE/ACM 5th International Workshop on Data Analysis and Reduction for Big Scientific Data (DRBSD-5)* 2019
- [85] Authors not available. “Understanding Performance-Quality Trade-offs in Scientific Visualization Workflows with Lossy Compression,” *2019 IEEE/ACM 5th International Workshop on Data Analysis and Reduction for Big Scientific Data (DRBSD-5)* 2019
- [86] Authors not available. “Virtual and Augmented Reality for Visualization of Big Data: Examples from Deep Earth to Subsurface,” *AGU Fall Meeting Abstracts* 2019
- [87] Authors not available. “A view from ORNL: Scientific data research opportunities in the big data age,” *2018 IEEE 38th International Conference on Distributed Computing Systems (ICDCS)* 2018
- [88] Authors not available. “Binning Based Data Reduction for Vector Field Data of a Particle-In-Cell Fusion Simulation,” *International Conference on High Performance Computing* 2018
- [89] Authors not available. “Coupling exascale multiphysics applications: Methods and lessons learned,” *2018 IEEE 14th International Conference on e-Science (e-Science)* 2018
- [90] Authors not available. “Dense Texture Flow Visualization using Data-Parallel Primitives.,” *EGPGV@ EuroVis* 2018
- [91] Authors not available. “Exploring Earth’s interior in collaborative immersive VR environments,” *AGU Fall Meeting Abstracts* 2018

- [92] Authors not available. “*Exploring Earth’s interior in collaborative immersive VR environments,*” *AGU Fall Meeting 2018* 2018
- [93] Authors not available. “*Global adjoint tomography-new generation earth mantle model,*” *AGU Fall Meeting Abstracts* 2018
- [94] Authors not available. “*In situ analysis and visualization of fusion simulations: Lessons learned,*” *International Conference on High Performance Computing* 2018
- [95] Authors not available. “*Canopus: A paradigm shift towards elastic extreme-scale data analytics on hpc storage,*” *2017 IEEE International Conference on Cluster Computing (CLUSTER)* 2017
- [96] Authors not available. “*Canopus: Enabling extreme-scale data analytics on big $\{\$HPC\}$ storage via progressive refactoring,*” *9th $\{\$USENIX\}$ Workshop on Hot Topics in Storage and File Systems (HotStorage 17)* 2017
- [97] Authors not available. “*Exacution: Enhancing scientific data management for exascale,*” *2017 IEEE 37th International Conference on Distributed Computing Systems (ICDCS)* 2017
- [98] Authors not available. “*Visualising Earth’s Mantle based on Global Adjoint Tomography,*” *AGU Fall Meeting Abstracts* 2017
- [99] Authors not available. “*Stream processing for near real-time scientific data analysis,*” *2016 New York Scientific Data Summit (NYSDS)* 2016
- [100] Authors not available. “*Tuned to terrible: A study of parallel particle advection state of the practice,*” *2016 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)* 2016
- [101] Authors not available. “*Visualization and analysis for near-real-time decision making in distributed workflows,*” *2016 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)* 2016
- [102] Authors not available. “*Visualization and analysis requirements for in situ processing for a large-scale fusion simulation code,*” *2016 Second Workshop on In Situ Infrastructures for Enabling Extreme-Scale Analysis and Visualization (ISAV)* 2016
- [103] Authors not available. “*Loosely coupled in situ visualization: A perspective on why it’s here to stay,*” *Proceedings of the First Workshop on In Situ Infrastructures for Enabling Extreme-Scale Analysis and Visualization* 2015
- [104] Authors not available. “*Towards scalable visualization plugins for data staging workflows,*” *Big Data Analytics: Challenges and Opportunities (BDAC-14) Workshop at Supercomputing Conference* 2014
- [105] Authors not available. “*Adios visualization schema: A first step towards improving interdisciplinary collaboration in high performance computing,*” *2013 IEEE 9th International Conference on e-Science* 2013
- [106] Authors not available. “*Computation and volume rendering of large-scale EOF coherent modes in rotating turbulent flow data,*” *AGU Fall Meeting Abstracts* 2013

- [107] Authors not available. “A distributed data-parallel framework for analysis and visualization algorithm development,” *Proceedings of the 5th Annual Workshop on General Purpose Processing with Graphics Processing Units* 2012
- [108] Authors not available. “Collaborative monitoring and visualization of HPC data,” *2012 International Conference on Collaboration Technologies and Systems (CTS)* 2012
- [109] Authors not available. “Mining hidden mixture context with ADIOS-P to improve predictive pre-fetcher accuracy,” *2012 IEEE 8th International Conference on E-Science* 2012
- [110] Authors not available. “Parallel stream surface computation for large data sets,” *Ieee symposium on large data analysis and visualization (ldav)* 2012
- [111] Authors not available. “Pattern Detection and Extreme Value Analysis on Large Climate Data,” *AGU Fall Meeting Abstracts* 2011
- [112] Authors not available. “A case study of collaborative facilities use in engineering design,” *The Engineering Reality of Virtual Reality 2010* 2010
- [113] Authors not available. “Occam’s razor and petascale visual data analysis,” *Journal of Physics: Conference Series* 2009
- [114] Authors not available. “Scalable computation of streamlines on very large datasets,” *Proceedings of the Conference on High Performance Computing Networking, Storage and Analysis* 2009
- [115] Authors not available. “Adsorption and decomposition of decamethylferrocene on Ag (100).,” *ABSTRACTS OF PAPERS OF THE AMERICAN CHEMICAL SOCIETY* 2000
- [116] Authors not available. “Adsorption of nickelocene on Si (111).,” *ABSTRACTS OF PAPERS OF THE AMERICAN CHEMICAL SOCIETY* 2000
- [117] Authors not available. “Visual interfaces for solid modeling,” *UIST* 1995
- [118] Authors not available. “Visual interfaces for solids modeling,” *Proceedings of the 8th annual ACM symposium on User interface and software technology* 1995

Workshop Papers

- [119] Moreland, K., Pugmire, D., Chen, J.. “The Exploitation of Data Reduction for Visualization,” *Unknown Venue* 2022 DOI: [10.2172/1881112](https://doi.org/10.2172/1881112).

Book Chapters

- [120] Pugmire, D., Ross, C., Thompson, N., Kress, J., Atkins, C., Klasky, S., Geveci, B.. “Fides: A General Purpose Data Model Library for Streaming Data,” *Lecture Notes in Computer Science* 2021 DOI: [10.1007/978-3-030-90539-2_34](https://doi.org/10.1007/978-3-030-90539-2_34).
- [121] Authors not available. “In Situ Visualization of Radiation Transport Geometry,” *Proceedings of the In Situ Infrastructures on Enabling Extreme-Scale Analysis and Visualization* 2017

- [122] Authors not available. *“Pillars of the Mantle: Imaging the Interior of the Earth with Adjoint Tomography,” Proceedings of the Practice and Experience in Advanced Research Computing 2017 on Sustainability, Success and Impact* 2017
- [123] Authors not available. *“Improving Cache Energy Efficiency for Green Computing,” Encyclopedia of Business Analytics and Optimization* 2014
- [124] Authors not available. *“Magnetic field outflows from active galactic nuclei,” Proceedings of the 2011 companion on High Performance Computing Networking, Storage and Analysis Companion* 2011

Other Publications

- [125] Moreland, K., Pugmire, D., Geveci, B., Lo, L., Childs, H., Bolstad, M., Shah, R., Wu, P.. *“The Importance of Scientific Visualization on Novel Hardware,” Unknown Venue* 2022 DOI: [10.2172/1885308](https://doi.org/10.2172/1885308).
- [126] Authors not available. *“Parallel Particle Advection Bake-Off for Scientific Visualization Workloads,” Unknown Venue* 2020
- [127] Authors not available. *“A Lifeline-Based Approach for Work Requesting and Parallel Particle Advection,” Unknown Venue* 2019
- [128] Authors not available. *“In situ particle advection via parallelizing over particles,” Unknown Venue* 2019
- [129] Authors not available. *“PAVE: An In Situ Framework for Scientific Visualization and Machine Learning Coupling,” Unknown Venue* 2019
- [130] Authors not available. *“ATDM/ECP Milestone Memo WBS 2.3. 4.04/SNL ATDM Data and Visualization Projects STDV04-21-[MS1/YR2] Q3: Prototype Catalyst/ParaView in-situ viz for unsteady RV flow on ATS-1,” Unknown Venue* 2018
- [131] Authors not available. *“In Situ Visualization for Computational Science,” Unknown Venue* 2018
- [132] Authors not available. *“October 2018 ECP ST Project Review: ECP Project WBS 2.3. 4.13 VTK-m/ECP.,” Unknown Venue* 2018
- [133] Authors not available. *“Performance-portable particle advection with vtk-m,” Unknown Venue* 2018
- [134] Authors not available. *“Canopus: A Paradigm Shift Towards Elastic Extreme-Scale Data Analytics on HPC,” Unknown Venue* 2017
- [135] Authors not available. *“ECP Milestone Report WBS 2.3. 4.13 ECP/VTK-m FY18Q1 [MS-18/01-03] Multiblock/Gradients/Release STDA05-5.,” Unknown Venue* 2017
- [136] Authors not available. *“ECP/VTK-m (ECP-ST Summary Slide).,” Unknown Venue* 2017

- [137] Authors not available. *“Pillars of the Mantle: Imaging the Interior of the Earth with Adjoint Tomography. In: PEARC17 Proceedings of the Practice and Experience in Advanced Research Computing 2017 on Sustainability, Success and Impact, Article No. 75,” Unknown Venue 2017*
- [138] Authors not available. *“In situ terminology project,” Unknown Venue 2016*
- [139] Authors not available. *“Performance modeling of in situ rendering,” Unknown Venue 2016*
- [140] Authors not available. *“XVis: Visualization for the Extreme-Scale Scientific-Computation Ecosystem: Year-end report FY16 Q4.,” Unknown Venue 2016*
- [141] Authors not available. *“A data centric view of large-scale seismic imaging workflows,” Unknown Venue 2013*
- [142] Authors not available. *“High performance visualization: Enabling extreme-scale scientific insight,” Unknown Venue 2012*
- [143] Authors not available. *“Hybrid parallelism,” Unknown Venue 2012*
- [144] Authors not available. *“Mining Hidden Mixture Context with ADIOS-P to Improve Predictive Pre-fetcher Accuracy,” Unknown Venue 2012*
- [145] Authors not available. *“Parallel Integral Curves.,” Unknown Venue 2012*
- [146] Authors not available. *“New insights into the oxidation/corrosion of plutonium,” Unknown Venue 2011*
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- [148] Authors not available. *“VISMASHUP: streamlining the creation of custom visualization applications,” Unknown Venue 2010*
- [149] Authors not available. *“Seeing the Unseeable,” Unknown Venue 2008*

Awards

- Award Name, Organization, Year.
- Another Award, Organization, Year.

Community Service

- Program Chair, IEEE VIS 2025/2026.
- General Chair, IEEE Uncertainty Visualization Workshops 2024/2025.